A Comparative Study of Statistical learning Methods for Classifying Cancer Subtypes Using Gene Expression Data

Dissertation Submitted in Partial Fulfilment of the
Requirements for the degree of

MASTER OF SCIENCE

By

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May 2025

Certificate

This is to certify that Dipmalya Roy, a student of M.Sc. in Statistics (2023–2025), has carried out the Research Project entitled **"A Comparative Study of Statistical learning Methods for Classifying Cancer Subtypes Using Gene Expression Data"** under the guidance of Dr. Aniket Biswas.

To the best of our knowledge, the work submitted is original and has not been carried out elsewhere in any other University or Institute.

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Declaration

I hereby declare that the project work entitled **"A Comparative Study of Statistical learning Methods for Classifying Cancer Subtypes Using Gene Expression Data"** submitted in partial fulfilment of the requirements for the degree of M.Sc. in Statistics (2023–2025) is my original work.

I affirm that, to the best of my knowledge, I have given due credit and cited all sources of information used in the preparation of this project. No part of this work has been plagiarized or submitted elsewhere for the award of any other degree or diploma.

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Semester : IV, M.Sc. Statistics

2023-2025

Acknowledgement

I would like to express my heartfelt gratitude to all those who have supported and guided me in the successful completion of this project. First and foremost, I wish to convey my profound gratitude and deepest regards to Dr. Aniket Biswas, St. Xavier's University, Kolkata, for his unwavering support, insightful guidance, valuable feedback, and encouragement throughout the course of this project. His vision and mentorship have been instrumental in shaping this work. I am also sincerely thankful to Prof. Manisha Pal, Dr. Priyanka Talukdar, Dr. Utpal Pore for their invaluable insights, thoughtful suggestions, and academic direction, which have greatly contributed to the depth and quality of this study. Finally, I extend my sincere thanks to St. Xavier's University, Kolkata for providing the necessary resources, infrastructure, and a conducive academic environment that enabled me to carry out this project to the best of my ability.

Dipmalya Roy

Abstract

This dissertation investigates into the application of RNA-Sequence data, a rich source of information about gene activity, and statistical learning techniques to the challenging problem of classifying multiple cancer types. My primary focus is on distinguishing between three prevalent and well-studied cancers: Breast (BRCA), Lung (LUAD), and Colorectal (COAD). RNA-Sequence technology, with its ability to provide a comprehensive view of the transcriptome, allows us to identify gene expression patterns that are uniquely characteristic of different cancer subtypes. This research details the methodology I employed, encompassing several key steps: data preprocessing to ensure quality, dimensionality reduction using Principal Component Analysis (PCA) , cluster analysis to group the co-expressed genes , feature selection with Elastic Net regularization to pinpoint the most relevant genes, and the application of a suite of classification algorithms: Support Vector Machines (SVM), Random Forest, K-Nearest Neighbours (KNN), and Gaussian Discriminant Analysis (GDA). Notably, this study also incorporates feature importance estimation using the Random Forest algorithm, providing insights into the specific genes that most significantly contribute to the classification of these cancer types. The performance of these models is rigorously evaluated using Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) and Accuracy. The empirical findings illustrate the high discriminative power of the statistical models applied in this work for the RNA-sequence dataset considered, underscoring the importance of appropriate modelling in in advancing cancer diagnostics and paving the way toward more individualized therapeutic approaches. This work contributes meaningfully to the ongoing effort to understand and combat cancer.

Keywords: RNA sequence, Cancer diagnostics, Elastic Net regularization, Classification

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Chapter 1: Introduction

1.1. Background

Cancer, in its diverse forms, remains a significant global health challenge. It's not a single disease, but rather a complex group of diseases characterized by the fundamental disruption of normal cellular processes. This disruption leads to uncontrolled cell growth, where cells divide and proliferate without the usual checks and balances.(Huang et al., 2024) A particularly sinister aspect of cancer is its ability to invade nearby tissues and spread to distant sites in the body, a process known as metastasis. This complexity at the cellular level translates to a tremendous challenge in the clinic.

Accurate classification of cancer is fundamental to effective clinical management. It is essential for pinpointing the specific type and subtype of the malignancy, which in turn guides the determination of the disease's extent through staging and grading. This classification is also critical for predicting patient outcomes or prognosis and for selecting the most appropriate treatment strategies. Furthermore, precise cancer classification forms the bedrock of personalized medicine, an approach that aims to tailor therapies based on the unique molecular characteristics of an individual's cancer, ultimately striving for improved patient outcomes. Recognizing the inherent heterogeneity of cancer and treating each patient based on their specific disease profile is paramount for maximizing therapeutic efficacy

Traditional methods of cancer classification have historically relied heavily on histopathological analysis. This involves the microscopic examination of tissue samples by trained pathologists, who look for abnormalities in cell structure and organization. For many years, this approach has been the cornerstone of cancer diagnosis, and it has provided invaluable information. However, while histopathology has served us well, it also has limitations. It primarily focuses on the morphological characteristics of the tumour, and it may not always be able to fully capture the intricate molecular heterogeneity that exists within and between tumours. In other words, tumours that look similar under the microscope may, in fact, be very different at the molecular level, and this difference can have profound implications for how they behave and how they respond to treatment.

1.2. RNA Sequencing: Unveiling the Transcriptome

RNA sequencing is a sophisticated technique that allows for the examination of the quantity and sequences of RNA molecules within a biological sample through the use

of next-generation sequencing (NGS). A typical RNA-Sequence experiment begins with the isolation of RNA from the chosen biological material, such as cells or tissues. Ensuring that the extracted RNA is of sufficient quality is crucial for the success of the subsequent steps. Following isolation, specific subsets of RNA molecules may be selected using particular protocols, for instance, poly-A selection to enrich for messenger RNA (mRNA) transcripts or ribo-depletion to remove ribosomal RNAs (rRNA).The isolated RNA is then reverse-transcribed into complementary DNA (cDNA), a more stable form suitable for sequencing(Kukurba & Montgomery, 2015) .The cDNA molecules are subsequently fragmented, and sequencing adapters, which are short DNA sequences necessary for the sequencing process, are ligated to the ends of these fragments.9 In some instances, a polymerase chain reaction (PCR) amplification step is performed to increase the quantity of the prepared sequencing library. The resulting library, containing millions of cDNA fragments with attached adapters, is then sequenced on an NGS platform.(Kukurba & Montgomery, 2015).This sequencing process generates millions of short nucleotide sequences, known as 'reads', which originate from random positions along the input RNA molecules .These reads, typically ranging from 30 to 400 base pairs in length depending on the sequencing technology used, are the primary output of the sequencing run .Following sequencing, these reads undergo a computational process called alignment, where they are mapped back to a reference genome or a reference transcriptome. In cases where a reference genome is not available, the reads can be assembled de novo, meaning they are assembled into a transcriptome without prior genomic information. This entire process, from RNA isolation to the generation of aligned reads, provides a detailed and quantitative representation of the transcriptome of the sample at the time of RNA extraction

1.3. RNA Sequence Data and Cancer Classification

Traditional morphological methods for cancer classification have significant limitations, particularly in capturing the molecular heterogeneity of underlying cancer subtypes. In response to these challenges, **RNA sequencing** has emerged as a powerful tool for comprehensive transcriptome profiling. RNA-Sequence utilizes high-throughput sequencing technologies to provide a detailed and quantitative representation of the transcriptome—the complete set of RNA molecules transcribed in a cell or tissue at a given time. Compared to earlier methods such as Sanger sequencing and microarray-based platforms, RNA-Sequence offers markedly higher coverage, resolution, and sensitivity (Wang et al., 2009). It enables the accurate quantification of gene expression levels, identification of alternative splicing events, and detection of allele-specific expression.

Recently, RNA-Sequence has been successfully able to classify subtypes in various other cancers. In **colorectal cancer**, gene expression profiling via RNA-Sequence has led to the identification of consensus molecular subtypes (CMS), which have clear

biological interpretability and hold promise for guiding clinical stratification and the development of subtype-based targeted interventions. RNA-Sequence has also been applied to classify other cancers, such as glioblastoma, hepatobiliary cancer, and pancreatic cancer, based on their unique gene expression profiles, contributing to a more nuanced understanding of these complex diseases.[41] These diverse examples underscore the broad utility of RNA-Sequence in oncology, highlighting its ability to refine cancer classification across a multitude of tumour types.

1.4. Challenges with RNA Sequence Data

In typical RNA-Sequence experiments, the expression levels of tens of thousands of genes are measured, while the number of samples remains relatively small. This results in a setting where the number of features (p) far exceeds the number of observations (n), creating a "High dimensional setup", thus creating following significant challenge in statistical analysis.

- **Curse of Dimensionality**: When of features (p) far exceeds the number of observations (n) methods like Ordinary Least Squares (OLS) regression become unstable or infeasible because the matrix that needs to be inverted is either singular or nearly singular. Since design matrix become rank-deficient and its inverse doesn't exist. Thus, there is no unique solution to estimation of parameter leading to instability in parameter estimation.

  As dimensionality increases, data points become sparse and increasingly distant from each other. This sparsity reduces the effectiveness of many statistical and machine learning algorithms, especially those relying on distance or density estimation.(Fan et al., 2014)

- **Computational Burden**: High-dimensional matrices are large and dense, making operations such as matrix inversion or eigen decomposition computationally expensive and numerically unstable.

- **Spurious Correlations and Overfitting**: A spurious correlation is a statistical relationship between two variables that appears to be causal or meaningful, but is actually driven by some external factor, confounding variable, or random chance. It occurs when two variables are correlated due to a third, unseen variable, or purely due to the limited sample size and randomness in high-dimensional data. In high-dimensional settings, spurious correlations are particularly common due to the large number of variables relative to the number of samples, leading to false associations. With thousands of features, it becomes easy to find seemingly strong associations purely by chance. This increases the risk of overfitting, where the model captures noise with respective to specific sample rather than true underlying patterns, leading to poor generalization to new data. Thus, the model overfit. (Fan et al., 2014)

1.5. Literature Review

Recent advances in transcriptomic technologies have greatly enhanced the ability to classify and understand cancer subtypes through RNA-Sequence data. RNA-Sequence provides a comprehensive and sensitive approach for quantifying gene expression profile, offering advantages over earlier technologies like microarrays (Wang et al., 2009). However, challenges such as technical noise and high dimensionality necessitate careful normalization and preprocessing. A 2023 investigation by Ahlmann-Eltze and Huber compared various transformation techniques for RNA-Sequence data, emphasizing their impact on data quality and downstream analyses(Ahlmann-Eltze & Huber, 2023). Lytal in 2020 conducted an empirical survey on normalization methods for single-cell RNA-Sequence, demonstrating that method choice significantly affects clustering and classification outcomes(Lytal et al., 2020). Similarly, Luecken and Theis (2019) provided a tutorial on best practices for single-cell RNA-Sequence analysis, stressing that appropriate normalization is essential for interpreting biological signals(Luecken & Theis, 2019). Zararsız et al. (2017) further contributed by proposing tissue-aware normalization techniques to manage sample heterogeneity, which is especially relevant in cancer studies(Zararsız et al., 2017). Koch et al. (2018) used transcriptomic profiling to identify developmental trajectories and heterogeneity in melanoma(Koch et al., 2018), while Qi et al. (2020) leveraged AI-based analysis for predicting outcomes in non-small cell lung cancer(Qi et al., 2020). Zararsız (2017) further exhibits different classification techniques (Zararsız et al., 2017) . A study by Huang (2024) reviewed the landscape of bioinformatics tools for cancer applications, noting the rapid integration of machine learning and RNA-Sequence analytics.(Huang et al., 2024) In the domain of cancer subtype discovery, Collectively, these studies underscore the importance of integrating robust statistical preprocessing with modern computational techniques to harness the full potential of RNA-Sequence for cancer research.

1.6. Data Source

The gene expression dataset that has been utilized in this study was obtained from the UCI Machine Learning Repository (Fiorini, 2016). This dataset, titled "gene expression cancer RNA-Seq", comprises of RNA sequencing (RNA-Seq) measurements of 20,531 different gene expression for 801 samples across 5 types of cancer namely: Breast (BRAC), Colorectal (COAD), Kidney (KIRC), Lung (LUAD), Prostate (PRAD); measured by illumina HiSeq 2000 next generation sequencing techniques. The data is publicly available under the Creative Commons CC0 license and can be accessed via the provided DOI: https://doi.org/10.24432/C5R88H

1.7 Objectives of the Study

The primary objective of this study is to explore and develop a comprehensive statistical learning framework for the multi-cancer classification from RNA-Sequence gene expression data. Specifically, the study focuses on five major cancer types: Breast Invasive Carcinoma (BRCA), Lung Adenocarcinoma (LUAD), Colon Adenocarcinoma (COAD), Kidney Renal Clear Cell Carcinoma (KIRC), and Prostate Adenocarcinoma (PRAD). Specific objective includes

(i) Performing rigorous preprocessing and normalization of high-dimensional RNA-Sequence gene expression datasets.

(ii) Identification of groups of genes with similar expression pattern.

(iii) Application and Evaluation of the performance achieved by various machine learning classifiers—including Support Vector Machines (SVM), Random Forest (RF), K-Nearest Neighbours (KNN), and Gaussian Discriminant Analysis (GDA)—for accurate classification of the cancer subtypes.

(iv) Identification of plausible biomarker genes.

An additional objective of this study is to construct an end-to-end statistical learning strategy tailored to the complexity and dimensionality of transcriptomic data. This includes both algorithmic evaluation and interpretability aspects, with a strong emphasis on biological interpretability.

1.8. Methodology

This dissertation presents a comprehensive analytical framework for addressing the challenge of analysis of RNA Sequence datasets. The methodological pipeline begins with an extensive data preprocessing phase, which includes logarithmic transformation ($\log_2$), scaling, and variance filtering. These steps are crucial for stabilizing variance, normalizing the data distribution, and removing low-information features, thereby enhancing the quality and suitability of the dataset for downstream analyses.

Following preprocessing, the analysis progress along two parallel methodological branches. The first branch involves dimensionality reduction through Principal Component Analysis (PCA), aimed at capturing the most informative components. Subsequently, unsupervised clustering techniques—specifically k-means and k-medoids—are applied to this reduced feature space to uncover latent structure and natural groupings within the data.

The second branch emphasizes supervised learning and begins with feature selection using the Elastic Net regularization method, which balances the strengths of both L1

and L2 penalties to identify a subset of the most informative genes. These selected features are then utilized in multiple classification algorithms, including Support Vector Machines (SVM), Random Forests, k-Nearest Neighbours (KNN), and Gaussian Discriminant Analysis (GDA). This enables both classification and estimation of feature importance.

Finally, results from both branches—unsupervised clustering and supervised classification—are integrated in a concluding synthesis phase. This integrative approach facilitates a holistic understanding of both the underlying data structure and its predictive characteristics, advancing the potential for accurate diagnostics, biomarker discovery, and targeted therapeutic strategies.

1.9. Outline

The structure of this dissertation is organized into seven chapters, each addressing a critical component of the multi-cancer classification pipeline using RNA-Sequence data. The second chapter focuses on data preprocessing. The third chapter is dedicated to the cluster analysis, where unsupervised learning techniques are applied to uncover latent structures within the data. The fourth chapter deals with feature selection, specifically through the use of Elastic Net regularization, to identify a subset of informative genes which are relevant for classification tasks. Chapter five presents the classification stage, where multiple supervised learning algorithms are employed to build predictive models for distinguishing between cancer types. The sixth chapter is concerned with the extraction of important genes, emphasizing the identification of biologically meaningful features with potential diagnostic or therapeutic relevance. Finally, the seventh chapter offers concluding remarks and outlines directions for future work, reflecting on the findings and limitations of the study.

Chapter 2: Data Preprocessing

2.1. Data Description and Visualization

Here we have 20,533 genes and 801 samples and no missing values. The distribution of different types of cancer in data as follows:

- **Table No: 2.1**

A Table summarizing the number of samples per cancer type

Cancer Type	Abbreviation	Number of Samples
Breast	BRAC	300
Colorectal	COAD	78
Kidney	KIRC	146
Lung	LUAD	141
Prostate	PRAD	136

Since the most frequent class BRAC is 300 out of 801, hence it is approximately 37 percent of the total sample size. So, we can conclude that the dataset is not imbalance and need no specialized treatment for typical imbalanced data.

We use a heatmap to visualize gene expression profile. It gives us an idea that only a handful of genes exhibits variability across different biological condition. So, it can be concluded from here that the data set is full of noise and only very few genes possess useful information. This process led us to use variance filtering in downstream analysis.

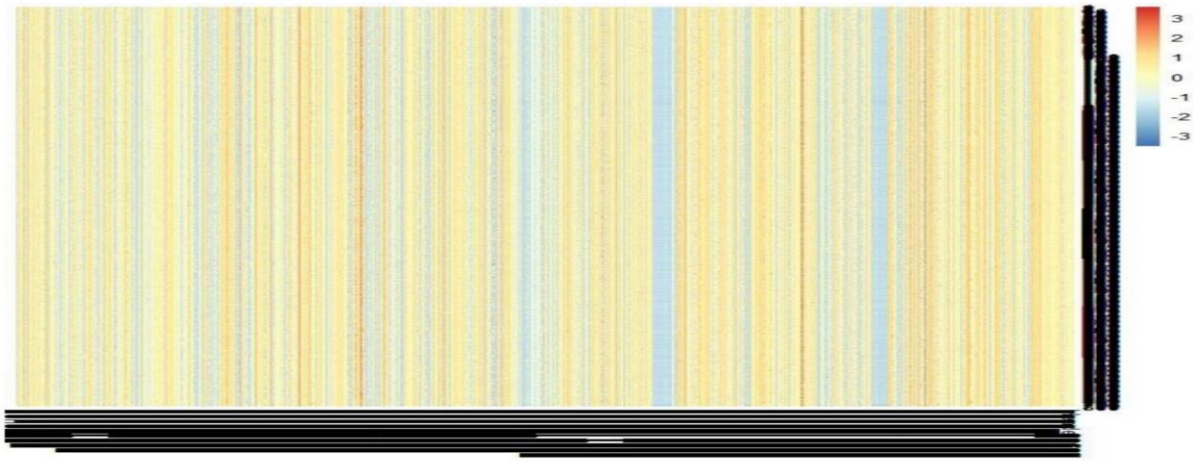


Fig 2.1: A heatmap showing the expression level of genes

2.2. Data Preprocessing

Raw RNA-Sequence data can be noisy and heteroscedastic, requiring preprocessing to ensure data quality and prepare it for downstream analysis. Though there are different schools of thoughts (Ahlmann-Eltze & Huber, 2023; Lytal et al., 2020) .For the purposes of this Study, the following methodological approach was implemented.

2.2.1 Normalization

- log_2 transformation: Apply log_2 (x + 0.25) to stabilize variance and handle zero values. (where represent the gene expression level.)
- Scaling: Scale the log_2 transformed data to have zero mean and unit variance.

- **Justification of Logarithmic Transformation: -**

- **Addressing Heteroscedasticity and Stabilizing Variance:** RNA Sequence data possess heteroscedasticity (Chatterjee et al., 2024)that violates the assumptions of downstream Statistical analysis. The log_2 transformation effectively compresses the scale of the data, particularly for highly expressed genes. This compression reduces the spread of the data points at higher expression levels, thereby stabilizing the variance near zero. By making the variance more uniform, the log-transformed data better satisfies the assumptions of statistical analysis, leading to more reliable and accurate results.

- **Normalizing Skewness and Approximating Normality:** RNA-sequence count data often follows a skewed distribution often negative binomial or poisson distribution as only few genes are highly correlated with any phenotypes(Church et al., 2019). This leads to some adverse effect on downstream statistical analysis such as high variance (Extreme values disrupt the consistency of variance, thus leading to

heteroscedasticity), misleading outliers that disproportionately influence the analysis and misspecification of model. Thus the log_2 transformation compress the the right tail of the distribution, shifting the data to a more symmetrical form that closely approximatea normal distribution. While RNA-sequence data hardly transformed to perfectly normal after log transformation, it significantly reduces skewness, making the data more suitable to statistical analysis.

- **Biological Significance:** - In the field of Biology while studying the gene expression we measure the ratio of expression level for two different conditions, this is called fold change. Raw RNA sequence data are often measured on additive scale, but the biological reality a change from 10 to 20 (2-fold) is not same as change from 100 to 110 (1.1-fold), hence directly applying any additive method will be not appropriate for biological interpretation. So, this transformation allows us to analyse expression changes on an additive scale that directly interpretable on multiplicative scale

$$\log(\text{Fold Change}) = \log\left(\frac{A}{B}\right) = \log(A) - \log(B)$$

Therefore, due to the above justification log transformation is widely employed in the preprocessing of RNA sequencing data. Accordingly, we also selected log transformation to preprocess our data.

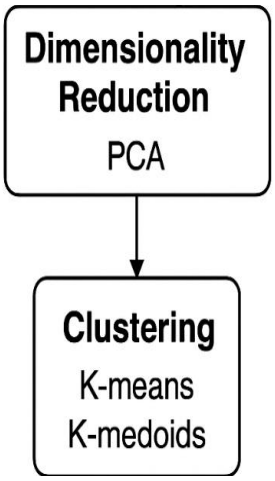
2.2.2. Variance Filtering

One common approach in Filtering method is called Variance filtering where we removed the genes which have low variance across the different sample under the assumption that the genes which shows hardly any variance under different biological condition are highly unlikely to differentially expressed or carry information for phenomenon under investigation. Thus, genes which have low variance are removed for which we used a data driven approach rather than an arbitrary one. We have selected a threshold of 80 percentile as we removed all genes exhibits variance bellow this particular threshold, thus selecting 4107 genes. Hence this step remove noise while preserving the necessary information. Also, Variance filtering reduce the burden of Multiple testing.(Bourgon et al., 2010)

Hence, the dataset has been appropriately pre-processed and is now ready for subsequent statistical analysis. In the following sections, we proceed with clustering and classification to uncover underlying patterns and predictive structures within the data.

Chapter 3: Cluster Analysis

To identify distinct groups of genes exhibiting similar expression patterns across the experimental conditions, clustering analysis was performed on the previously filtered set of the top 20% most variable genes**.** But clustering in high-dimensional set up poses significant statistical challenges, mainly due to the sparsity of data as dimensionality increases as we encounter a phenomenon known as “Curse of dimensionality”. In such scenario, distance matrix become less informative, as the relative differences between pairwise distances diminish, making it increasingly difficult to distinguish between clusters. To address this issue, we implemented the following dimensionality reduction strategy prior to clustering by principal component analysis and then procced with clustering analysis.



3.1. Dimension Reduction: Principal Component Analysis

3.1.1 Methodology

Principal Component Analysis is a data reduction technique; where we want to explain the variance-covariance matrix of data by linear combination of the data itself, keeping in mind of reduction of dimensionality and orthogonality of dimension. Let originally our data has P dimension; now let we take K linear combination of these Variables; (where K < P). Now this new K dimension observations has same variability of the original data set. Hence dimensionality reduction is achieved. In our study dimensionality reduction has two main objectives Visualization and Summarization. For Visualization the objective to consider the most optimal two or three dimensions and use them as coordinates to get a visual representation of the data, but for summarization it is prescribed to consider the essential components that sufficiently capture the inherent information of the data. For our study the number of components is selected by Scree Plot.

Scree Plot is a diagrammatic mechanism to determine the appropriate number of principal components. With the eigenvalues ordered in increasing order of magnitude, we plot magnitude of eigenvalues against its number. To determine the number of components we consider the elbow bend in the diagram as the remaining eigenvalues hardy add any information about the data.

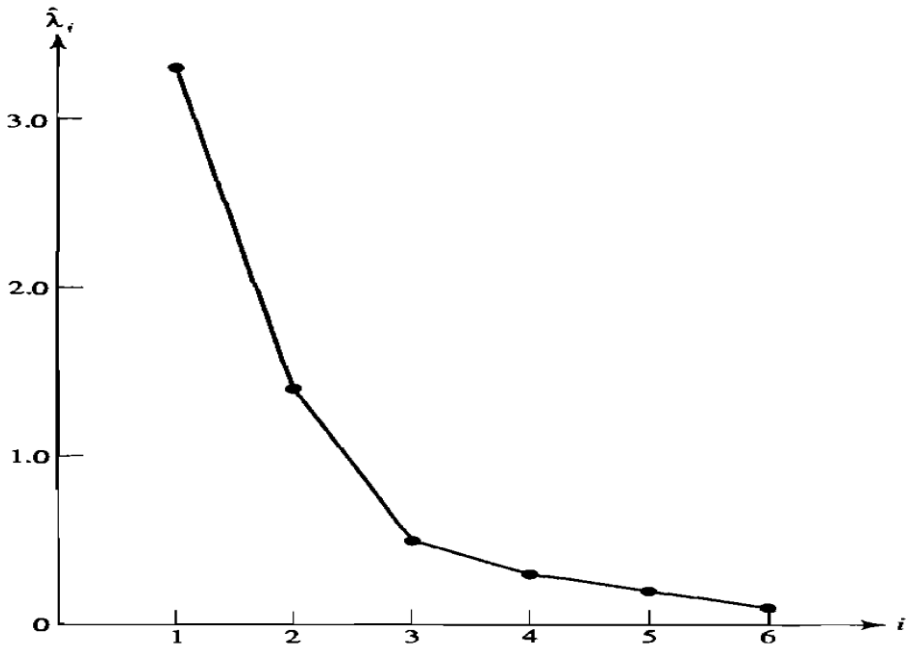


Fig 3.1: A representative image of Scree Plot

3.1.2 Results

Five principal components have been selected by the scree plot (Fig: 3.1.2(a)) as since five it exhibits a diminish return. The first two components are considered as coordinates for visualization of a PCA cluster (Fig 3.1.2(b)).

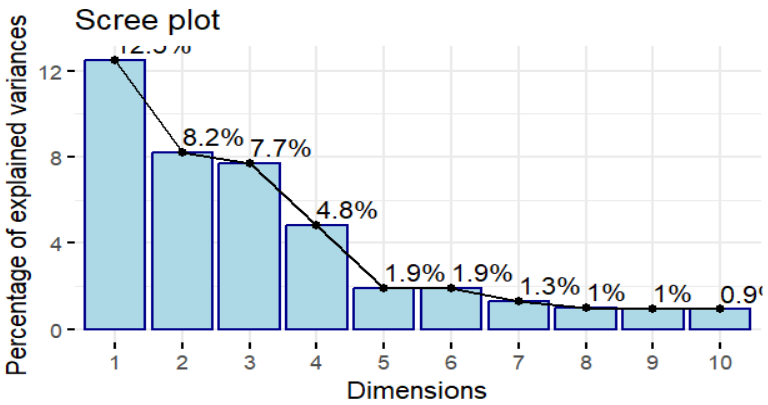


Fig 3.3.1(a): Scree Plot

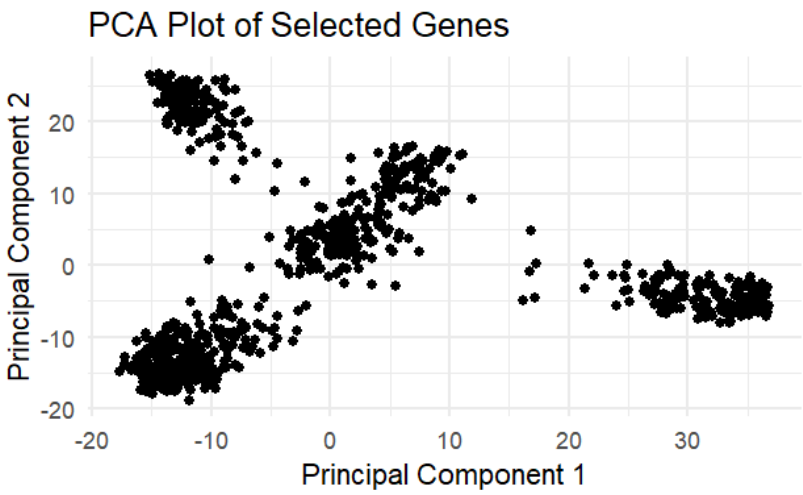


Fig 3.3.1(b): PCA Cluster

3.2. K -means Clustering

3.2.1. Methodology

K-means clustering which is a widely used partitioning algorithm, was applied to the reduced dimensional space defined by the selected principal components. In this approach we assign observations to the cluster so the mean of the clusters is as different as possible. Here we choose the initial cluster centre and assign each subject to its nearest cluster in terms of distance to the centroids. Then we find the new centroids based on this new cluster and again assign the points based on the same metric in terms of the new centroids. We repeat this whole process until it converges.

The algorithm intends to partition the genes into k distinct, non-overlapping clusters by iteratively assigning each gene to the cluster with the nearest centroid and then updating the centroids based on the mean of the genes assigned to each cluster.

The optimal number of clusters (k) was determined using the following methods:

- **Elbow Methodes**: As the clustering intent to minimize the within cluster variance and maximize the between cluster variance. Hence, we plot the magnitude of within-cluster sum of squares (WSS) values against the number of clusters and identifying the "elbow" point from where the rate of decrease in WSS starts to diminish.

- The stability of the clustering algorithm was tested by running the algorithm 25 times with different initialization.

3.2.2. Results

Employing twenty-five random start the appropriate number of clusters is determined to be five from elbow method (Fig 3.2.2(a)). When partitioning the data into five cluster, the between cluster sum of squares, explain 89% of total sum of square. Figure 3.2.2(b) provides a visual representation of the clustering results.

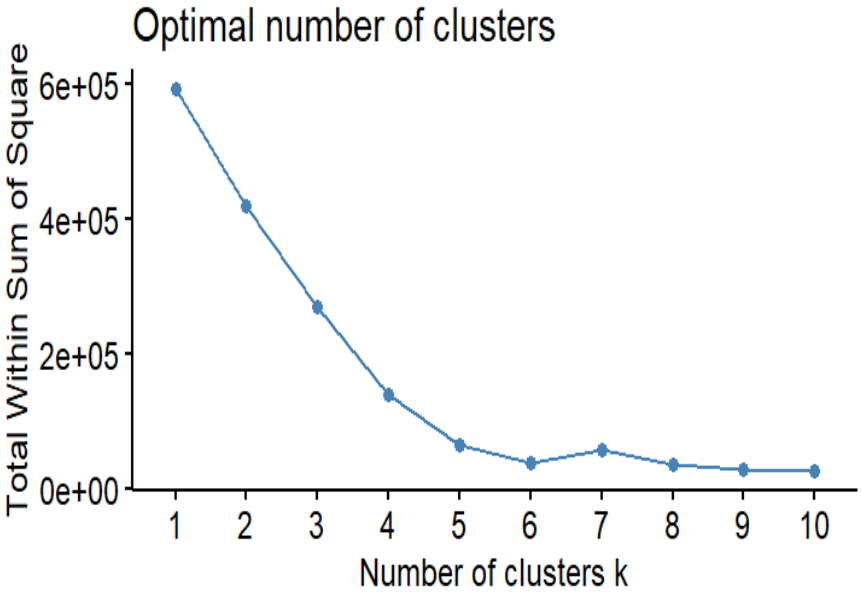


Fig 3.3.2(a): Elbow Method

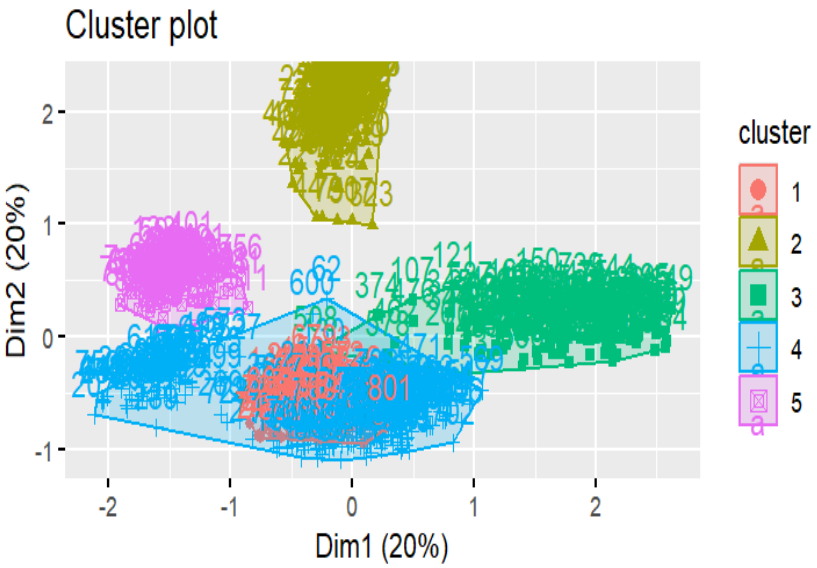


Fig 3.3.2(b): Visualization of 5 cluster

3.3. K-medoids Clustering

3.3.1. Methodology

To assess the robustness of the clustering results and to address the sensitivity of K-means to outliers, K-medoids clustering has been also performed. Unlike K-means, which uses the mean of the cluster members as the centroid, K-medoids selects actual data points (Medoids: The medoid of a cluster is formally defined as the data point within the cluster for which the total (or equivalently, the mean) dissimilarity to all other points in the cluster is minimized. In other words, it represents the most centrally located element with respect to pairwise dissimilarities. Unlike certain objects used by other algorithms, the medoid is an actual point in the cluster.) as representative of each cluster. This makes K-medoids less sensitive to extreme values. Like previous method we also select the number of clusters by elbow method.

Initialization:
Select k representative data points, referred to as medoids, from the total of n observations using a greedy strategy that seeks to minimize the initial clustering cost.

Assignment Phase:
Allocate each observations to the closest medoid based on a specified dissimilarity measure.

Optimization Loop (Swap Procedure):

Continue iterating while the total clustering cost decreases:

For each current medoid m and each non-medoid data point o, evaluate the change in total cost that would result from swapping mmm with o.

- Identify and record the pair ($m*$, $o*$) that yields the maximum cost reduction.
- If this optimal swap leads to a decrease in the objective function, perform the swap. Otherwise, terminate the algorithm.

3.3.2. Results

K-medoids clustering is also performed employing twenty-five random start and the number of clusters is selected by elbow method. The clustering results exhibits almost similarity with K-means Clustering is proof of effectiveness of our transformation steps as the influence of outliers have been effectively diminished.

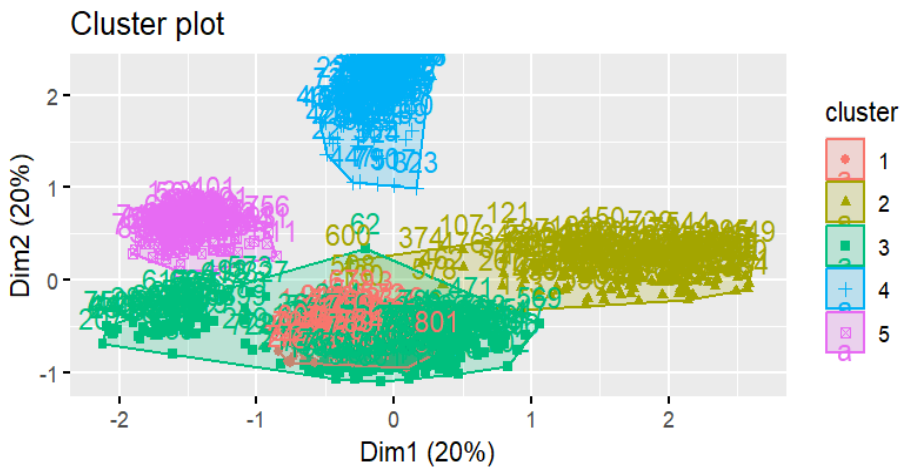


Fig 3.3.2: Visualization of 5 K-medoids cluster

3.4. Biological Interpretation

In this study, unsupervised clustering of gene expression profiles results in identification of five distinct gene clusters without any prior knowledge of the cancer types, has independently rediscovered the five known classes. Biologically, results of clustering suggests that the underlying molecular landscape of the cancer samples represented by five major gene expression patterns. Each cluster likely represents a set of co-regulated genes that participate in similar biological condition that relevant to neoplasia. The emergence of five cluster also reflects the heterogeneity of cancer subtype studied, with each cluster potentially corresponds to different oncological pathways.

Also, limited number of clusters, indicate that despite the complexity of the transcriptome data, a relatively small number of biological signatures that controls the gene expression variability across cancer subtypes. This insight creates a foundation for further analysis of function of each gene cluster and identification of plausible biomarkers and thus provides a pathway for development of targeted therapeutic strategies.

Chapter 4: Feature Selection

As the RNA sequence data set have more features than samples it falls under High Dimensional Statistical Data set and traditional approaches fell as it results in high variance and overfitting. Hence, we have to do a select a subset of features to overcome the curse of dimensionality. Here we use Elastic net (Zou & Hastie, 2005) to select a subset of feature.

4.1. Elastic Net

4.1.1 Methodology

After Variance Filtering Elastic Net regularization was employed for further feature selection. This method combines the L1 (Lasso) and L2 (Ridge) penalties, offering benefits from both. The techniques follow as:(Zou & Hastie, 2005)

- Setup: Consider a dataset with n observations with p covariates where p>>n.
- Let y= $[y_{1,}y_{2,}.....,y_n]^T$ be the response and X=[x_1 | x_2 | … |x_p] be the model matrix then For any fixed non-negative λ_1 and λ_2 we define the naive elastic net criterion as :

$$L(\lambda_1, \lambda_2, \beta) = |\mathbf{y} - \mathbf{X}\beta|^2 + \lambda_2|\beta|^2 + \lambda_1|\beta|_1,$$

Where

$$|\beta|^2 = \sum_{j=1}^{p} \beta_j^2,$$

$$|\beta|_1 = \sum_{j=1}^{p} |\beta_j|.$$

The naïve Elastic net estimator of β is

$$\hat{\beta} = \arg\min_{\beta}\{L(\lambda_1, \lambda_2, \beta)\}.$$

It can be viewed as penalized least square methods. Let $\alpha = \frac{\lambda_1}{\lambda_1 + \lambda_2}$,then solving β in the above equation is equivalent to the optimization problem

$$\hat{\beta} = \arg\min_{\beta} |\mathbf{y} - \mathbf{X}\beta|^2, \qquad \text{subject to } (1-\alpha)|\beta|_1 + \alpha|\beta|^2 \leqslant t \text{ for some } t.$$

The constrain is a convex combination of the both Lasso and ridge penalty. When α=0 it become simply become Ridge regression and when it is 1 it behaves like Lasso. To find the optimal value of α we will use cross-validation.

- **Benefits of Elastic Net**
  - Grouped Selection: Once a predictor is selected within a highly correlated group the whole group is selected. Which is of paramount important for our study and other methods failed miserably in this criterion(Zou & Hastie, 2005)

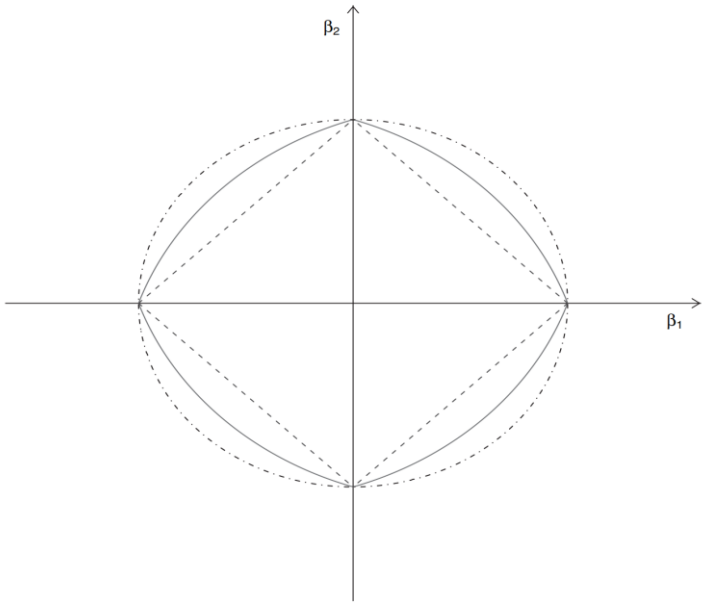


Fig 3.3.2: Two-dimensional contour plots (level 1) outer is ridging penalty and innermost lasso while the middle one is of Elastic net with mixing proportion α=0.05

4.1.2. Results

Elastic Net regression was applied to the selected genes using a regularization parameter of α=0.1 and a penalty term λ=0.0801, obtained through 10-fold cross-validation using "accuracy" as a metrics. This procedure resulted in the selection of 680 genes for further analysis.

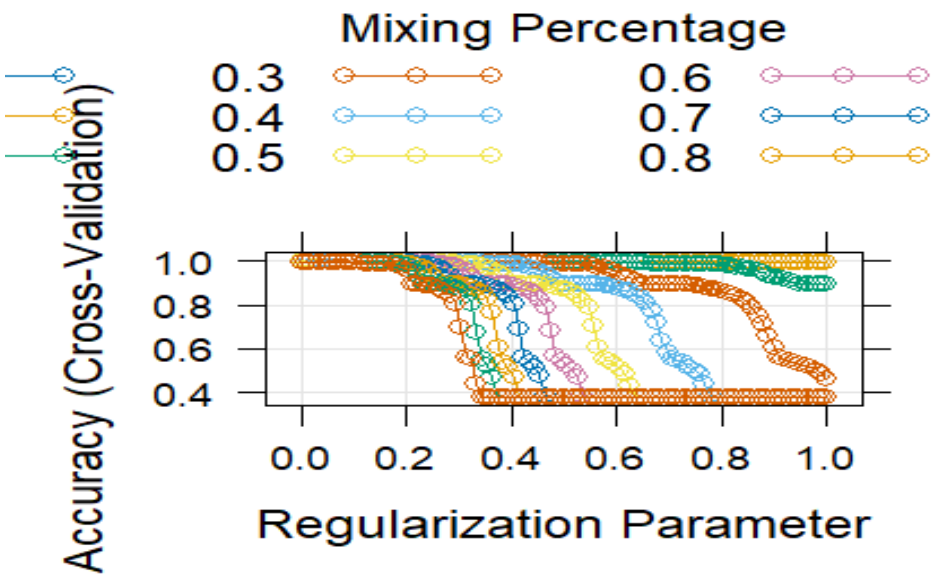


Fig 4.1.2: Visualization of feature selection by Elastic net

Chapter 5: Classification

Classification techniques machine learning algorithms that learn from labelled data, by training on a dataset where the cancer subtype for each sample is already known. The goal of these methods is to build a predictive model that can accurately classify new, unlabelled cancer samples into one of the predefined subtypes**.**

5.1. Data splits and hyperparameter Tunning

To ensure the reliability and robustness of the classification models built using supervised methods, rigorous training and validation strategies are essential**.** A common strategy that has been employed in our study is splitting the data into a training set (80 % of total), used to train the model, and a separate testing set (remaining 20 %), used to evaluate the model's ability to accurately classify the samples it has never encountered before, to assess how well the model is likely to perform when applied to new, independent datasets.

Now to find the optimal hyperparameter for best model performance the method of k fold cross validation technique was used. The core idea is to partition the training data set in k equally sized folds. The model is trained in k-1 fold and validated on the remaining fold on turn. We select the hyperparameter that results in best accuracy from a grid of hyperparameters.

5.2 The Classification Models

According to the literature review and their suitability for RNA sequencing data set, we selected the following four models for evaluation: **Support Vector Machine (SVM) with Radial Basis Function (RBF) kernel**, **Random Forest**, **K-Nearest Neighbours (KNN)**, and **Gaussian Discriminant Analysis (GDA)**. These models represent a diverse set of classification approaches, balancing between complexity, interpretability, and adaptability to high-dimensional gene expression data. The **SVM with RBF kernel** is effective in capturing non-linear relationships (soft margin) and is particularly well-suited for datasets with many features relative to observations. **Random Forest** is a robust ensemble method that handles overfitting and variable importance estimation efficiently. **KNN**, being a non-parametric method, is simple yet powerful, while **GDA** provides a probabilistic framework assuming multivariate Gaussian distributions, often useful in biological classification problems. All models were subjected to **K-fold cross-validation for hyperparameter tuning**, ensuring optimal performance and minimizing the risk of overfitting due to the high dimensionality of RNA-sequence data.

5.3 Support vector Machine with RVF kernel

5.3.1 Methodology

Support Vector Machines (SVMs) is a supervised classification algorithms that at its core aim to find the optimal hyperplane that separates data points in n dimensions of different classes with the largest possible margin (distance of points from decision boundary). An SVM with a radial basis function (RBF) kernel was used to model potential non-linear relationships between gene expression and cancer type.

Where RBF kernel is defined as K (x, x') = exp (−γ ||x − x'||²). where a small γ means decision boundary is smooth and less sensitive to individual point hence may underfit and large γ may tends to overfit.

The optimization function in its primal form is

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i$$
$$\text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i$$

The dual problem is advantageous while using the kernel

$$\max_{\boldsymbol{\alpha}} \quad \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{n}\alpha_i\alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$$
$$\text{subject to} \quad 0 \leq \alpha_i \leq C, \quad \sum_{i=1}^{n}\alpha_i y_i = 0$$

where k (x_i ,x_j) is a kernel function.

Now the decision function is as follows

After solving for α_i, predict with:

$$f(\mathbf{x}) = \sum_{i=1}^{n}\alpha_i y_i \exp\left(-\gamma\|\mathbf{x}_i - \mathbf{x}\|^2\right) + b$$

Classify on the basis of sign of f(x).

- Here C is the regularization parameter. Now a high value of C means the algorithm classify all the training example correctly potentially at the cost of very narrow margin (overfitting), while a small value of C may lead to underfitting.

So, to find the optimal value of the hyperparameter γ and C we use cross validation.

5.3.2. Support Vector Machine with RVF Kernel Training Results

Support Vector Machine (SVM) classifiers with a Radial Basis Function (RBF) kernel were applied to classify cancer subtypes based on RNA sequencing data comprising 680 samples and 640 predictors across five classes. The value σ = 0.001054207 was held constant. Accuracy was used to select the optimal model using the largest value using 5-fold cross validation. (Fig 5.3.2) The final values used for the model were sigma = 0.001054207 and C = 8.

The final model attained an average cross-validated accuracy of 82.8%, with an AUC of 0.997 and a Kappa statistic of 0.761.

Table No: 5.3.2: Performance Metrics of the Optimal SVM Model (5-fold Cross-Validation)

Metric	Value
Optimal C	8
Sigma (RBF Kernel Width)	0.0010
Accuracy	82.8%
Kappa	0.7610
Log Loss	0.4046
AUC	0.9973
Precision (Mean)	93.96%
Recall (Mean)	80.5%
F1 Score (Mean)	78.9%
Balanced Accuracy	87.5%
Sensitivity (Mean)	80.5%
Specificity (Mean)	94.5%

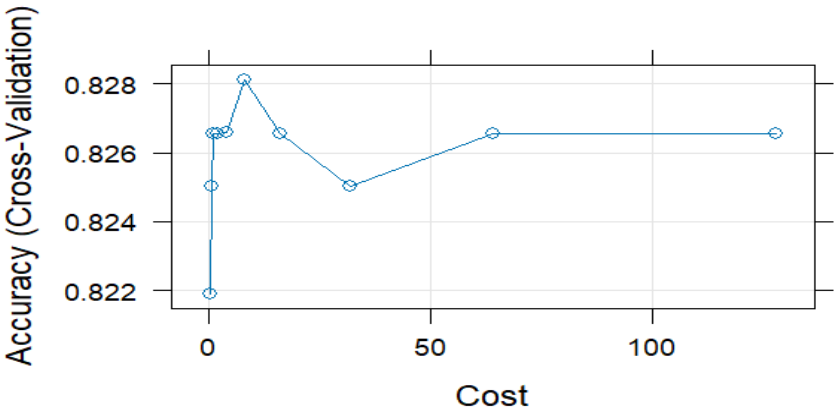


Fig: 5.3.2: Visualization of performance across set of hyperparameter

5.3.3. Performance on Test data:

- Following shows the confusion matrix

Actual \ Predicted	BRCA	COAD	KIRC	LUAD	PRAD
BRCA	**57**	**0**	**0**	**28**	**0**
COAD	**0**	**19**	**0**	**0**	**0**
KIRC	**0**	**0**	**27**	**0**	**0**
LUAD	**0**	**0**	**0**	**0**	**0**
PRAD	**0**	**0**	**0**	**0**	**30**

The following table summarizes the key metrics for the SVM model based on the confusion matrix results.

Table No: 5.3.3: Performance Metrics of the SVM Model

Metric	Class: BRCA	Class: COAD	Class: KIRC	Class: LUAD	Class: PRAD
Sensitivity	1.0000	1.0000	1.0000	0.0000	1.0000
Specificity	0.7308	1.0000	1.0000	1.0000	1.0000
Pos Pred Value	0.6706	1.0000	1.0000	Nan	1.0000
Neg Pred Value	1.0000	1.0000	1.0000	0.8261	1.0000
Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Rate	0.3540	0.1180	0.1677	0.0000	0.1863
Detection Prevalence	0.5280	0.1180	0.1677	0.0000	0.1863
Balanced Accuracy	0.8654	1.0000	1.0000	0.5000	1.0000

Though the average prediction accuracy is 82.61 % but the model performs poorly in classifying lung cancer almost half the time misclassifying it.

5.4. Random Forest

5.4.1. Methodology

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions. For classification tasks, the output of the random forest is the class selected by maximum number of trees. Since its development by Tim Kam Ho in 1995, Random Forest algorithm achieved its popularity for its robustness and ability to handle high-dimensional data. Since it is prone to overfitting hence there is a need hyperparameter tunning. In this study we focus on mtry.(Probst et al., 2018)

5.4.2. Random Forest Classification Training Results

A Random Forest classifier was trained on the RNA sequencing dataset comprising 640 samples and 680 predictors, corresponding to five cancer subtypes Model evaluation was performed using 5-fold cross-validation. Across varying values of *mtry* (the number of predictors randomly selected at each decision split), the model demonstrated exceptionally high classification performance. The final optimal model was selected based on the highest cross-validated accuracy, corresponding to mtry = 97. The key performance metrics for the final model are summarised in Table 5.4.2

Table No: 5.4.2: Performance Metrics of the Optimal Random Forest Model (5-fold Cross-Validation)

Metric	Value
Optimal mtry	97
Accuracy	99.7%
Kappa	0.9960
Log Loss	0.0531
AUC	1.0000
Precision (Mean)	99.84%
Recall (Mean)	99.65%
F1 Score (Mean)	99.74%
Balanced Accuracy	99.78%
Sensitivity (Mean)	99.65%

Specificity (Mean)	99.90%

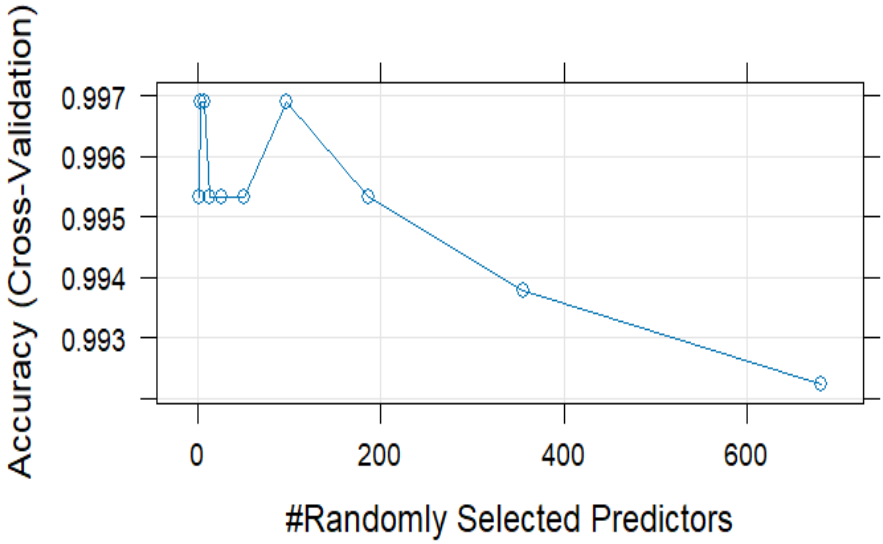


Fig: 5.4.2: Visualization of performance across set of hyperparameter

5.4.3 Performance on Test data:

- Following shows the confusion matrix

Actual \ Predicted	BRCA	COAD	KIRC	LUAD	PRAD
BRCA	57	0	0	0	0
COAD	0	19	0	0	0
KIRC	0	0	27	0	0
LUAD	0	0	0	28	0
PRAD	0	0	0	0	30

The following table shows the performance of random forest model on test data

Table No: 5.4.3: Performance Metrics of the SVM Model

Metric	Class: BRCA	Class: COAD	Class: KIRC	Class: LUAD	Class: PRAD
Sensitivity	1.0000	1.0000	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000	1.0000	1.0000
Pos Pred Value	1.0000	1.0000	1.0000	1.0000	1.0000
Neg Pred Value	1.0000	1.0000	1.0000	1.0000	1.0000
Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Rate	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Balanced Accuracy	1.0000	1.0000	1.0000	1.0000	1.0000

The model shows excellent performance across all the classes with 100 % accuracy rate of classification.

5.5. K- Nearest Neighbour Method (KNN)

5.5.1. Methodology

K-Nearest Neighbours (KNN) is a non-parametric method that classifies a new unseen sample based on plurality of its k nearest training point on feature space. The hyperparameter K is selected via Cross validation.

5.5.2. K-nearest Neighbour Training Results

The tuning process selected k=23 as the final model parameter. Accuracy has been used to choose the best model using the highest value based on the 5-fold cross-validated performance. The following table shows the metric of the final model.

Table No: 5.5.2: Performance Metrics of the Optimal KNN(K=23) Model (5-fold Cross-Validation)

Metric	Value
Accuracy	1.0000
Kappa	1.0000
Log Loss	0.0010
AUC (Area Under Curve)	1.0000
pacu (Precision-Recall AUC)	0.0034
Mean F1 Score	1.0000
Mean Sensitivity (Recall)	1.0000
Mean Specificity	1.0000
Mean Precision	1.0000
Mean Negative Predictive Value	1.0000
Mean Detection Rate	0.2000
Mean Balanced Accuracy	1.0000

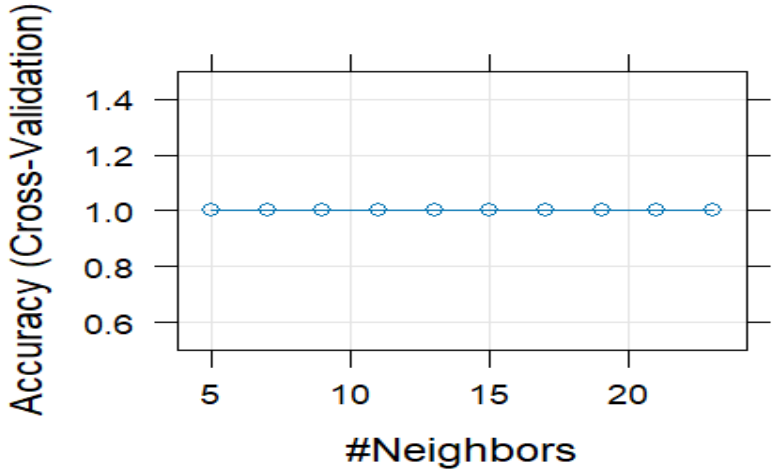


Fig: 5.5.2: Visualization of performance across set of hyperparameter

5.5.3. Performance on Test data

- The confusion matrix as follows

Actual \ Predicted	BRCA	COAD	KIRC	LUAD	PRAD
BRCA	57	0	0	0	0
COAD	0	19	0	0	0
KIRC	0	0	27	0	0
LUAD	0	0	0	28	0
PRAD	0	0	0	0	30

The following table summarizes the key metrics for the kNN model based on the confusion matrix results.

Table No: 5.5.3: Performance Metrics of the kNN Model

Metric	Class: BRCA	Class: COAD	Class: KIRC	Class: LUAD	Class: PRAD
Sensitivity	1.0000	1.0000	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000	1.0000	1.0000
Pos Pred Value	1.0000	1.0000	1.0000	1.0000	1.0000
Neg Pred Value	1.0000	1.0000	1.0000	1.0000	1.0000
Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Rate	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Balanced Accuracy	1.0000	1.0000	1.0000	1.0000	1.0000

The KNN algorithm shows excellent performance across all the classes with 100 % accuracy rate of classification.

5.6 Gaussian Discriminant Analysis (GDA)

5.6.1 Methodology

Gaussian Discriminant Analysis is a generative learning algorithm where we assume that each class follow a normal distribution and we put a new data point to a class for which we have maximum posterior probability.

5.6.2. **Gaussian Discriminant Analysis Training results**

The results of the Linear Discriminant Analysis (LDA) model exhibit exceptional classification performance in predicting cancer subtypes from RNA-Sequence data. The model achieved a perfect accuracy of 100%, with no misclassifications observed, as confirmed by the confusion matrix

Actual Predicted	BRCA	COAD	KIRC	LUAD	PRAD
BRCA	243	0	0	0	0
COAD	0	59	0	0	0
KIRC	0	0	119	0	0
LUAD	0	0	0	113	0
PRAD	0	0	0	0	106

The following table summarizes the key metrics for the Linear Discriminant Analysis (LDA) model based on the confusion matrix results.

Table No: 5.6.2: Summary of key metric

Metric	**Value**
Accuracy	1.0000
Kappa	1.0000
95% Confidence Interval for Accuracy	(0.9943, 1)
No Information Rate	0.3797
P-Value (Acc > NIR)	< 2.2e-16

Mcnemar's Test P-Value	NA
Balanced Accuracy (X1)	1.0000
Balanced Accuracy (X2)	1.0000
Balanced Accuracy (X3)	1.0000
Balanced Accuracy (X4)	1.0000
Balanced Accuracy (X5)	1.0000
Sensitivity (X1)	1.0000
Sensitivity (X2)	1.0000
Sensitivity (X3)	1.0000
Sensitivity (X4)	1.0000
Sensitivity (X5)	1.0000
Specificity (X1)	1.0000
Specificity (X2)	1.0000
Specificity (X3)	1.0000
Specificity (X4)	1.0000
Specificity (X5)	1.0000
Positive Predictive Value (X1)	1.0000
Positive Predictive Value (X2)	1.0000
Positive Predictive Value (X3)	1.0000
Positive Predictive Value (X4)	1.0000

Positive Predictive Value (X5)	1.0000
Negative Predictive Value (X1)	1.0000
Negative Predictive Value (X2)	1.0000
Negative Predictive Value (X3)	1.0000
Negative Predictive Value (X4)	1.0000
Negative Predictive Value (X5)	1.0000
Prevalence (X1)	0.3797
Prevalence (X2)	0.0921
Prevalence (X3)	0.1859
Prevalence (X4)	0.1766
Prevalence (X5)	0.1656
Detection Rate (X1)	0.3797
Detection Rate (X2)	0.0921
Detection Rate (X3)	0.1859
Detection Rate (X4)	0.1766
Detection Rate (X5)	0.1656
Detection Prevalence (X1)	0.3797
Detection Prevalence (X2)	0.0921
Detection Prevalence (X3)	0.1859
Detection Prevalence (X4)	0.1766

Detection Prevalence (X5)	0.1656

5.6.3. Performance on Test data

- The confusion matrix is as follows

Actual \ Predicted	BRCA	COAD	KIRC	LUAD	PRAD
BRCA	57	0	0	0	1
COAD	0	19	0	0	0
KIRC	0	0	27	0	0
LUAD	0	0	0	28	0
PRAD	0	0	0	0	29

The following table summarizes the key metrics for the GDA model based on the confusion matrix results.

Table No: 5.6.3: Performance Metrics of the SVM Model

Metric	Class: BRCA	Class: COAD	Class: KIRC	Class: LUAD	Class: PRAD
Sensitivity	1.0000	1.0000	1.0000	1.0000	0.9667
Specificity	0.9904	1.0000	1.0000	1.0000	1.0000

Metric	Class: BRCA	Class: COAD	Class: KIRC	Class: LUAD	Class: PRAD
Pos Pred Value	0.9828	1.0000	1.0000	1.0000	1.0000
Neg Pred Value	1.0000	1.0000	1.0000	1.0000	0.9924
Prevalence	0.3540	0.1180	0.1677	0.1739	0.1863
Detection Rate	0.3540	0.1180	0.1677	0.1739	0.1801
Detection Prevalence	0.3602	0.1180	0.1677	0.1739	0.1801
Balanced Accuracy	0.9952	1.0000	1.0000	1.0000	0.9833

Gaussian discriminant exhibits 99.38 % average accuracy across all classes of cancer while perfectly classifying colon, kidney and lung cancer.

5.7. Evaluation of Classification Model Performance

The performance of the previously decided classification models was evaluated using ROC curves and AUC score. The results are summarized in the table below:

Table 5.7.1: Summarization of AUC scores

Model	BRCA vs others	COAD vs others	KIRC vs others	LUAD vs others	PRAD vs others	Macro average
SVM with RVF kernel	1	1	1	0.9994	1	0.9998
Random Forest	1	1	1	1	1	1
K-nearest Neighbour	1	1	1	1	1	1
GDA	0.9951	1	1	1	1	0.9990

The ROC curves for each of the above model were generated, showing the trade-off between true positive rate and false positive rate. The AUC values indicate that all models performed exceptionally well in classifying the cancer types. Random Forest and K-Nearest Neighbours achieved perfect AUC scores of 1, while SVM and GDA also demonstrated very high performance.

Since the AUC scores are very close, we also look for accuracy of the models employed in our study.

Table 5.7.2: Summarization of Accuracy

Model	BRCA vs others	COAD vs others	KIRC vs others	LUAD vs others	PRAD vs others	Macro average
SVM with RVF kernel	86.54%	100%	100%	50%	100%	82.6%
Random Forest	100%	100%	100%	100%	100%	100%
K-nearest Neighbour	100%	100%	100%	100%	100%	100%
GDA	99.52%	100%	100%	100%	98.33%	99.38%

Chapter 6: Gene Feature Importance Extraction

In RNA-sequence dataset feature selection is an essential step to identifying key genes that contribute most significantly to classification performance not only enhances model efficiency but also enables the discovery of biologically relevant markers. Random Forest (RF), a widely used ensemble learning algorithm, is particularly well-suited for feature selection due to its inherent ability to estimate variable importance. This chapter focuses on leveraging the Random Forest algorithm to extract the most informative genes from RNA-sequence data, with the objective of improving biological interpretation.

6.1. Methodology: Feature Importance Estimation Using Mean Decrease in Impurity (MDI)

6.1.1. Node Impurity

Node impurity in a decision tree is a measurement of how mixed the class labels are within a node. One common impurity measure is the Gini index, defined for a node *t* as:

$$G(t) = 1 - \sum_{k=1}^{K} p_k^2$$

Where k is the number of classes and p_k is the proportion of class *k* observations in node t. When a feature *j* is used to split node *t* into left and right child nodes t_L and t_R the decrease in impurity due to feature j is

$$\Delta G_j(t) = G(t) - \left(\frac{N_{t_L}}{N_t} G(t_L) + \frac{N_{t_R}}{N_t} G(t_R) \right)$$

Where N_t is the number of samples in node t and G (t_L) and G(t_R) are Gini impurities of left and right nodes respectively.

6.1.2. Gini Impurity reduction by a Feature

The Gini impurity decrease for feature *j* across a single decision tree is the sum of the impurity reductions at every node *t* where feature *j* is used:

$$\text{ImpurityImportance}_j = \sum_{t \in T_j} \frac{N_t}{N} \Delta G_j(t)$$

where T_j is the set of nodes where feature *j* is used, and *N* is the total number of features.

6.1.3. Aggregating Importance Across the Forest

In a Random Forest, the importance of feature *j* is computed as the average impurity decrease across all trees:

$$\mathrm{MDI}_j = \frac{1}{T}\sum_{t=1}^{T}\mathrm{ImpurityImportance}_j^{(t)}$$

6.2. Results

The Random Forest model identified several genes with high importance scores, indicating a strong contribution in classification of cancer subtypes. The top 20 genes, ranked by their Gini importance, are visualized in the bar plot below:

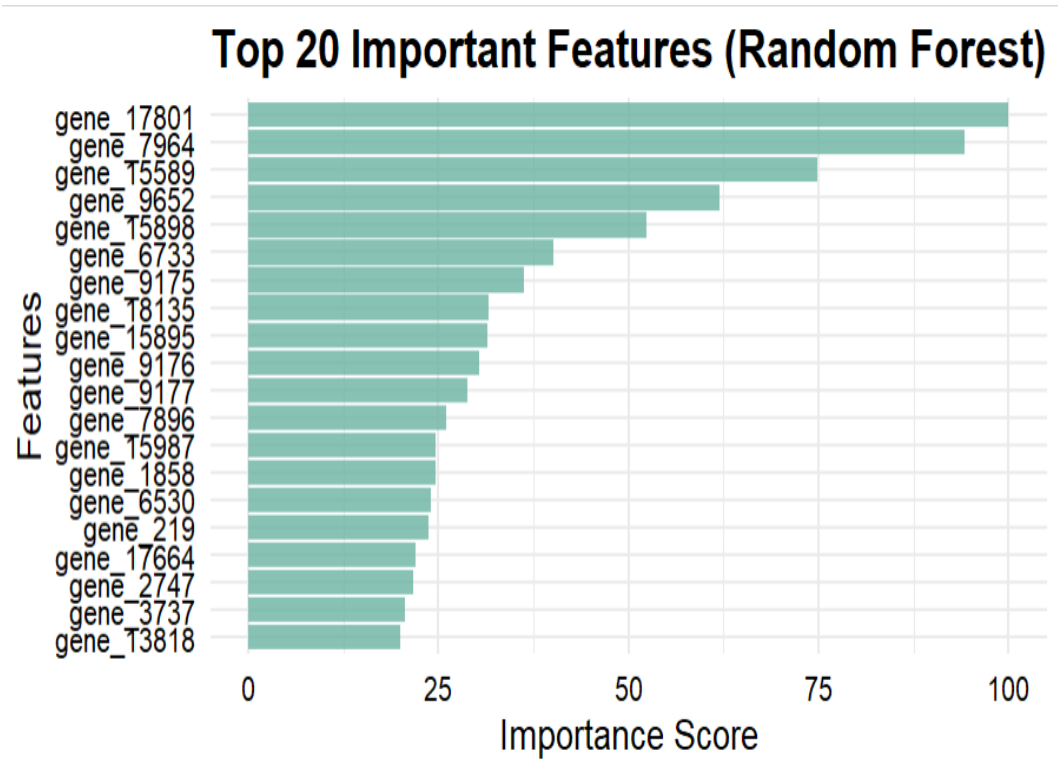


Figure 6.2: Top 20 Most Important Genes

The identification of these twenty most important genes provides valuable insights into the molecular mechanisms underlying disease states. These genes often emerge as plausible biomarkers with strong diagnostic and prognostic significance, as their expression patterns are consistently associated with specific clinical outcomes or phenotypic subgroups. As a result, they become ideal candidates for targeted therapeutic strategies. Such strategies may include traditional drug development aimed at modulating the activity of these genes, as well as advanced molecular approaches like gene editing using CRISPR-Cas9 systems. Additionally, RNA-based interventions—such as small interfering RNA (siRNA) and antisense oligonucleotides—offer promising avenues for post-transcriptional gene silencing, especially for genes that are difficult to target at the protein level. Therefore, the integration of statistical learning-based gene prioritization with molecular therapeutics holds significant potential for advancing precision medicine and improving patient-specific treatment outcomes.

Chapter 7: Future Works

This dissertation has demonstrated the efficacy of leveraging RNA-Sequence data in combination with statistical learning algorithms to achieve accurate multi-class classification across several major cancer types. By employing a structured pipeline—comprising data preprocessing, clustering, feature selection, classification, and gene importance extraction—this study has highlighted the potential of transcriptomic profiling as a powerful tool in modern cancer diagnostics. The results demonstrate the viability of data-driven approaches in identifying molecular patterns that distinguish cancer subtypes, contributing to advancements in personalized medicine and deepening our understanding of cancer biology. Looking ahead, this study opens up several promising avenues for further investigation. Future research may focus on the identification of differentially expressed genes to pinpoint novel therapeutic targets, as well as the validation of the current findings using independent external datasets. Additionally, expanding the analysis to encompass a broader spectrum of cancer types and integrating complementary omics data—such as genomics and proteomics—could substantially enhance the robustness and accuracy of cancer classification models. These directions hold promise for more comprehensive molecular characterization and improved clinical translation.

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